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SurvivalGPU: a GPU Powered Library for Scalable Survival Analysis in R and Python

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Introduction

SNDS: Social Security Data of 99% of the French population [1]

Adverse drug Reaction: can be analyzed using the WCE model[2,3]

Computation challenge: current implementation not adapted to large datasets in high dimension[4]

GPU acceleration: scale-up survival analysis with torch and pyKeops

Survival Analysis

Estimate the **time** before an event in the presence of censored data

The **Cox[6] model** is widely used

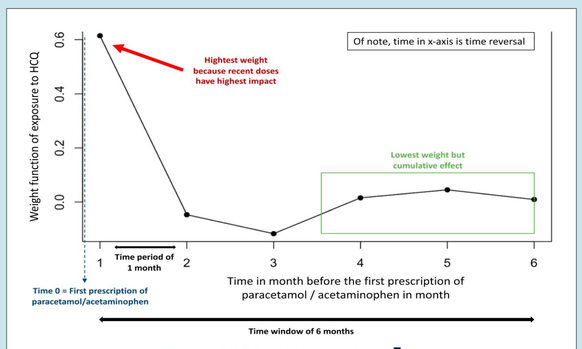
$$\mathbf{h(t; x_1, \dots, x_p) = h_0(t) e^{\beta_1 x_1 + \dots + \beta_p x_p}}$$

The coefficients are estimated by maximizing the **partial likelihood**

The WCE Model

The **WCE[3] function** represents the cumulative exposure as a **convolution**

$$\mathbf{WCE(u) = \sum_{t=1}^{u} w(u-t) X(t)}$$



The WCE function[5]

It is estimated as a linear combination of *m* cubic **B-splines**

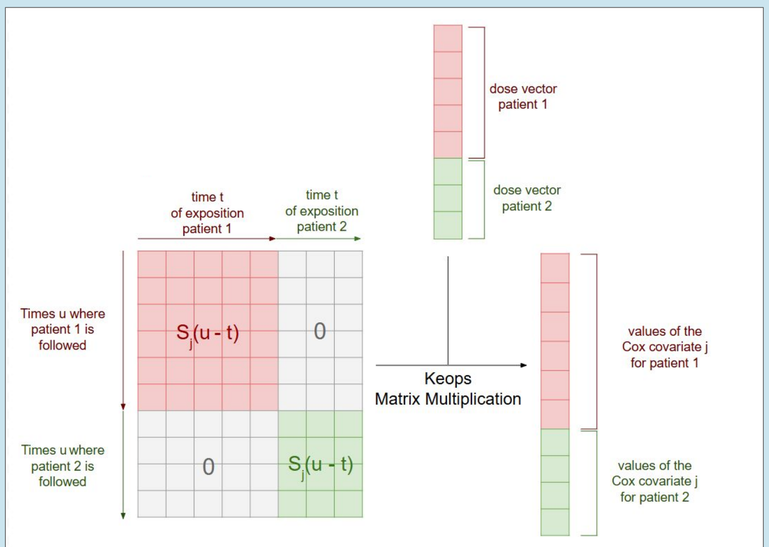
$$\mathbf{w(u-t) = \sum_{j=1}^{m} \theta_j B_j(u-t)}$$

and integrated in a **Cox Model**

$$\mathbf{D_j(u) = \sum_{t}^{u} B_j(u-t) X(t)} \quad \gamma_j = \beta\theta_j \longrightarrow \mathbf{h(u) = h_0(u) \exp\left[\sum_{j=1}^{m} \gamma_j D_j(u) + \sum_{s=1}^{q} \eta_s Z_s(u)\right]}$$

GPU Acceleration

The B-spline calculations used to compute the Cox covariates are GPU accelerated using **pyKeops[6,7]** for **memory optimization**



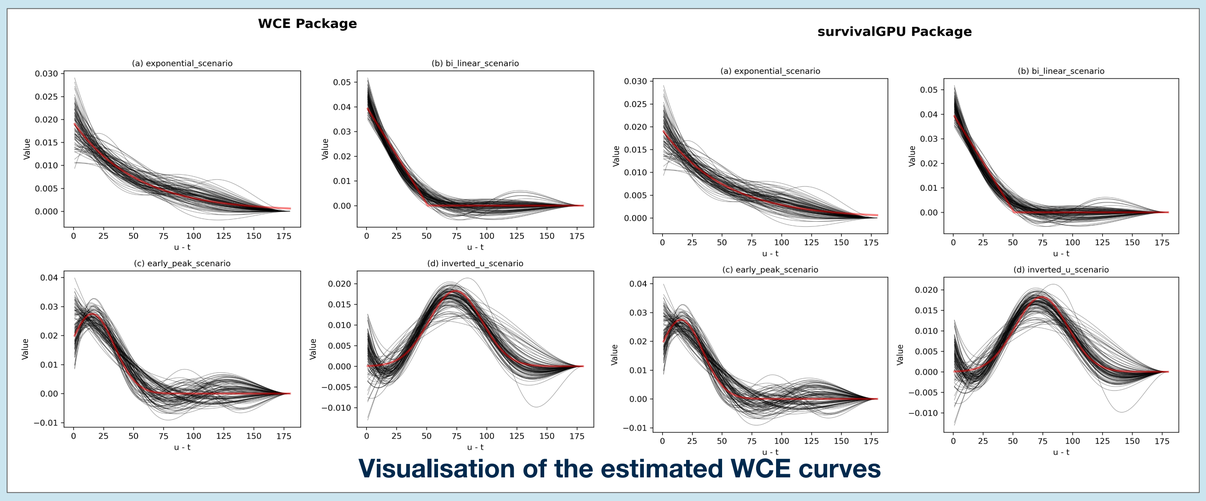
Calculation of Cox covariates for a single B-spline of formula S_j (example with two patients)

Package Validation

Comparison of **survivalGPU** and **WCE** implementations of the WCE model on simulated data with the **permalgo[3]**

		WCE package				survivalGPU			
Weight	True β	mean(β)	SD(β)	% Bias	Time (s)	mean(β)	SD(β)	% Bias	Time (s)
Exponential	1.386	1.351	0.207	-2.540	2.610	1.351	0.207	-2.540	2.080
Bi-linear	1.386	1.437	0.243	3.650	2.570	1.437	0.243	3.650	2.080
Early peak	1.386	1.349	0.196	-2.680	2.710	1.349	0.196	-2.670	2.080
Inverted U	1.386	1.369	0.230	-1.210	2.570	1.370	0.230	-1.200	2.070

Numerical comparison of the simulations

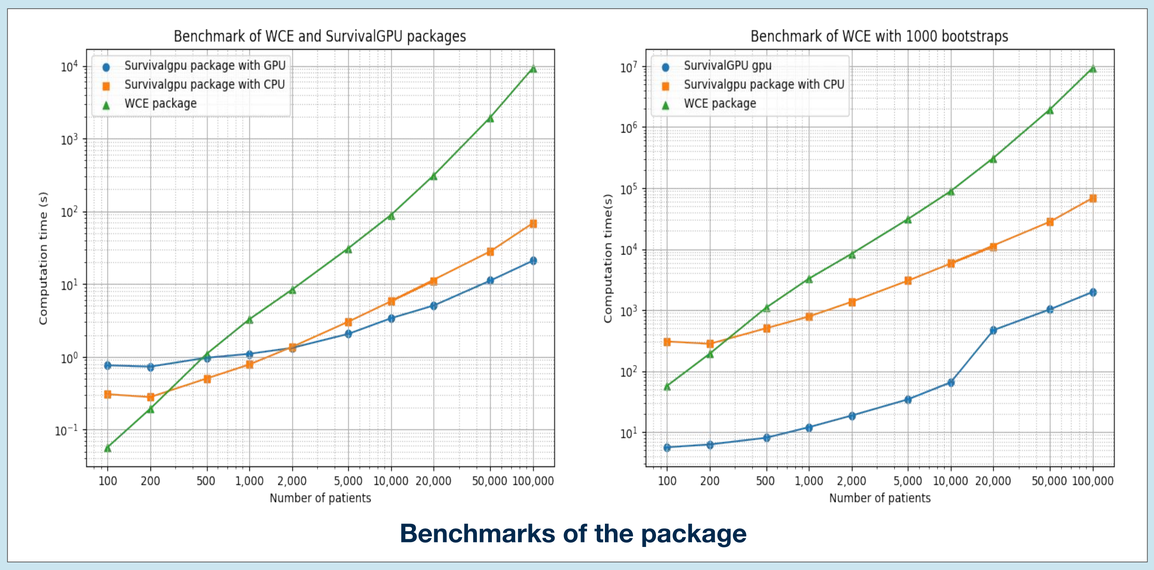


Visualisation of the estimated WCE curves

The WCE function of the two packages produce identical results
Our implementation is **validated**

Benchmarks

Datasets of **different sizes** are generated with the **permalgo[3]**



Benchmarks of the package

For large datasets our package is **100-1000 times faster**
The use of **GPU** is particularly relevant when using **bootstraps**

Conclusion

Validation: our package is equivalent to the R WCE package

Acceleration: high reduction of the computation time

High dimension: We will leverage GPU computation to extend the model to multiple drugs and interactions

Github

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[7] Charlier et al. (2021): *Kernel Operations on the GPU, with Autodiff, without Memory Overflows*